

First Term Re-Examination 2080

Class: XII 'Science'

Time: 3hr.

Subject: Chemistry [3021]

F.M: 75

P.M: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [11×1=11]

1. Equivalent weight of SO_2 in the reaction $\text{SO}_2 + \text{H}_2\text{S} \rightarrow \text{H}_2\text{O} + \text{S}$ is

- a) Mol.wt b) Mol.wt./2
c) Mol.wt/3 d) Mol.wt/4

2. The amount of NaOH required to neutralize 100 cm^3 of $0.1 \text{ M H}_2\text{SO}_4$ is

- a) 0.8g b) ~~0.4g~~ c) 1.2g d) 2.4g

- ### 3. Indicator is

- a) strong acid b) strong base
c) weak acid or weak base ~~c) weak organic acid or base~~

- 4. Common ion effect is not applicable for**

- a) HCl & H_2SO_4 b) H_2S & HCl
c) NH_4OH & NH_4Cl d) CH_3COOH & CH_3COONa

5. The compound whose 0.1M solution is basic is

- a) NH_4Cl
c) CHCOONH_4

6. The compound used as tear gas is

- a) $\text{Cl}_3\text{C}-\text{NO}_2$ b) $\text{CH}_3\text{CH}_2\text{OH}$
c) $\text{C}_6\text{H}_5\text{COCH}_3$ d) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$

- 7. An alkyl halide can be converted into alcohol by**

- a) Addition b) Elimination c) ~~Substitution~~ d) Dehydrogenation

8. The hybridization of carbon atom in haloarene is

- a) sp b) sp^2 c) sp^3 d) dsp^2

MCQs

Group 'A'

Chemistry

1. d. mol. wt / 4
2. a. 0.8g
3. c. weak acid or weak base
4. a. HCl & H₂SO₄
5. b. CH₃COONa
6. a. Cl₃C-NO₂
7. c. Substitution
8. b. sp²
9. b. Adsorption
10. c. +1 & +2
11. d. ZnCO₃

KMC Note

9. Froth floatation process is a practical example of

- a) Absorption
- b) Adsorption
- c) Coagulation
- d) Sedimentation

10. Common oxidation state of copper is

- a) +2 & +3
- b) +1 & +3
- c) +1 & +2
- d) +3 & +4

11. Calamine or Zinc spar has the formula

- a) ZnS
- b) ZnO
- c) Zn_2SiO_4
- d) ZnCO_3

GROUP 'B' [8×5=40]

Short Questions

1. a) Calculate the equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium with reaction. [2]

b) 0.14 g metal oxide is dissolved in 50 ml. seminormal hydrochloric acid. After the metal is dissolved solution is made half litre. 50 ml of this require $10 \text{ ml } \frac{N}{5}$ sodium hydroxide for neutralization. Calculate equivalent weight of metal oxide. [3]

2. a) Define Redox titration with suitable example. [2]

b) 10cc of N/2 HCl, 30cc of N/10 HNO_3 and 60cc of N/5 H_2SO_4 are mixed together. Calculate the pH of the mixture. [3]

3. a) Define ionic product of water. How does it depend upon temperature? [2]

b) Calculate the pH of $5 \times 10^{-8} \text{ M}$ HCl solution. [2]

c) Why is aqueous solution of CuSO_4 acidic in nature? [1]

4. a) Write short note on 'Chemistry of Blue vitriol'. [3]

b) What is blister copper? Explain. [2]

5. a) Explain why compounds of zinc are colourless? [1]

b) Mention the possible oxidation number of Mn. [1]

c) Most of the transition elements show paramagnetic behavior, why? [1]

d) Fe^{3+} is more stable than Fe^{2+} , why? [1]

- 11.
- 9) Explain why $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ ion is coloured? [1]
6. a) Write two general methods of preparation of iodoethane. [2]
- b) Convert 1-bromopropane to 2-bromopropane and vice versa. [3]
7. a) Write the reaction for preparation of trichloromethane from ethanol. [3]
- b) Give an example of: [1+1]
- i) Reimer – Tiemann's reaction
- ii) Williamson ether synthesis.
8. The nucleophilic substitution reactions are characteristics reactions of alkyl halide in which stronger nucleophile displaces a weaker nucleophile from substrate molecule.
- a) Why do alkyl halides undergo nucleophilic substitution reactions? [2]
- b) Starting from bromoethane how would you prepare
- i) ethanol ii) ethylethanoate
- iii) Diethyl thioether [3]

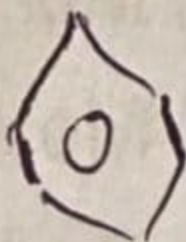
GROUP 'C' [3×8=24]

Long Questions

9. a) Define common ion effect with a suitable example. [2]
- b) Discuss about the application of common ion effect and solubility product principle. [3]
- c) The solubility product of CaCO_3 is 8.7×10^{-9} at 298K. Find its solubility in (a) pure water (b) 0.1M CaCl_2 . What mass of CaCO_3 is precipitated? [3]
10. a) Write the action of heat on white vitriol. [2]
- b) What happens when zinc is exposed to moist air? [1]
- c) What is lithopone? How is it formed? [1]
- d) Write down the process of extraction of zinc with necessary diagrams. [4]

11. a) A tired chemistry professor filled three bottles separately with primary, secondary and tertiary alcohols but forgot to label them. How would you distinguish them by Victor Meyer's method? [5]
- b) Chlorobenzene is less reactive than chloroethane towards nucleophilic substitution reaction, why? [2]
- c) Write short notes on "reduction of nitro benzene in various medium." [4]

"Best of Luck"



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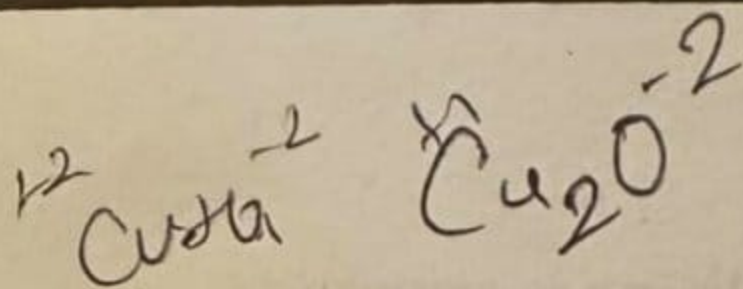
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Handwritten notes and calculations:

1000
6 (1)
3300
0.305

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